

Section Id :	64065324000
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	20
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065355946
Question Shuffling Allowed :	No

Question Number : 5 Question Id : 640653388702 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: INDUSTRY 4.0"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531291962. ✓ Yes

6406531291963. ✗ No

Sub-Section Number :	2
Sub-Section Id :	64065355947

Question Shuffling Allowed :

Yes

Question Number : 6 Question Id : 640653388709 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the correct progression of the industry?

Options :

6406531291972. ✖ Steam engine -> IT ->Assembly line ->IOT

6406531291973. ✖ Assembly line ->IT -> Steam engine ->IOT

6406531291974. ✖ Assembly line ->Steam engine -> IT ->IOT

6406531291975. ✔ Steam engine -> Assembly line ->IT ->IOT

Question Number : 7 Question Id : 640653388710 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which is the country of origin for Industry 4.0?

Options :

6406531291976. ✖ Japan

6406531291977. ✖ United States

6406531291978. ✖ Russia

6406531291979. ✔ United Kingdom

6406531291980. ✖ None of these

Question Number : 8 Question Id : 640653388711 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406531291981. ✖ IC engine

6406531291982. ✖ Electric car

6406531291983. ✔ Hyperloop

6406531291984. ✖ Steam locomotive

Question Number : 9 Question Id : 640653388712 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Location problem is not dependent on which of the following factors.

Options :

6406531291985. ✖ Distance

6406531291986. ✖ Number of facilities

6406531291987. ✖ Optimization criteria

6406531291988. ✔ None of these

Question Number : 10 Question Id : 640653388713 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the below formula represent the Euclidean metric distance?

Options :

6406531291989. ✔ $d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$

6406531291990. ✖ $d_{ij} = |(x_i - x_j)| + |(y_i - y_j)|$

6406531291991. ✖ $(x_2 - x_1) + (y_2 - y_1)$

6406531291992. ✖ All of these

Question Number : 11 Question Id : 640653388714 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

If we choose the **Euclidean** distance measure, what will the optimal solution be?

Options :

6406531291993. ✔ Euclidean distance measure will result in a single point optimal solution

6406531291994. ✖ Metropolitan metric distance will result in a range for the optimal solution

6406531291995. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are wrong

6406531291996. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are correct

Question Number : 12 Question Id : 640653388715 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

If we choose **Metropolitan distance** measure, how will the optimal solution be?

Options :

6406531291997. ✖ Euclidean distance measure will result in a single point optimal solution

6406531291998. ✔ Metropolitan metric distance will result in a range for the optimal solution

6406531291999. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are wrong

6406531292000. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are correct

Question Number : 13 Question Id : 640653388716 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Cross-median approach says that the location should be located at the median with respect to:

Options :

6406531292001. ✔ Demand density

6406531292002. ✖ Energy density

6406531292003. ✖ Both Demand density & Energy density

6406531292004. ✖ None of these

Sub-Section Number : 3

Sub-Section Id : 64065355948

Question Shuffling Allowed : No

Question Id : 640653388703 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (14 to 18)

Question Label : Comprehension

Currently, a company is using a layout which consists of 10 workstations. 200 cakes are baked in a day on this layout. A typical day consists of a single shift which runs for 8 hours. The cake baking process consists of 10 activities (the information is provided in the table below). Then answer the given subquestions

Activity ID	Activity Description	Activity Time (in seconds)	Preceding Activities
A	Prepare baking pans	20	
B	Make ingredients to room temperature	10	
C	Preheat Oven	60	
D	Stir ingredients together	40	A, B
E	Baking	20	C, D
F	Chocolate base pasting	5	E
G	Layer pasting	5	F, E
H	Icing	15	G
I	Cherry placing	5	H
J	Packing	5	I

Sub questions

Question Number : 14 Question Id : 640653388704 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the current efficiency of the line where each activity is performed on a separate workstation (enter only the numerical value without the "%" symbol. Example: if the answer is 99.756%" enter only "99.76")?

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

34.00 to 35.00

Question Number : 15 Question Id : 640653388705 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the maximum possible output that can be achieved?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

480

Question Number : 16 Question Id : 640653388706 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Given the activity information, (theoretically) how many workstations can be reduced from the current line while achieving the maximum possible output?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 17 Question Id : 640653388707 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which among the following layout is the best possible one to achieve the maximum possible output? (choose all that is applicable)

Options :

6406531291967. ✖ (A, B) → (C) → (D, E) → (F, G, H, I) → (J)

6406531291968. ✖ (A) → (C) → (B, E, F) → (D, G, H) → (I, J)

6406531291969. ✔ (A, B) → (C) → (D) → (E, F, G, H, I, J)

6406531291970. ✖ (A, B) → (C) → (D, G) → (E, F, H, I, J)

Question Number : 18 Question Id : 640653388708 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the efficiency of the best possible layout chosen in the previous question (entre only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” entre only “99.76”)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

77.0 to 77.3

Sub-Section Number : 4

Sub-Section Id : 64065355949

Question Shuffling Allowed : No

Question Id : 640653388717 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A

Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (19 to 22)

Question Label : Comprehension

Consider the following table, for the supplier selection problem. This table contains supplier attributes for 10 suppliers as explained in the tutorial. Answer the given subquestions

Supply network	Employee skills	Quality reputation	Price premium	Customers	Management
3	1	3	3	2	1
2	2	1	1	1	2
1	1	1	1	2	1
2	3	2	2	3	3
3	3	3	3	3	2
3	2	2	3	2	3
2	1	3	2	3	2
3	2	3	3	1	3
3	2	3	3	2	1
1	1	1	1	1	1

Sub questions

Question Number : 19 Question Id : 640653388718 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Multiple Choice Question

How many itemsets are there with cardinality/number of elements equal to 3?

Options :

6406531292005. ✖ 3060

6406531292006. ✖ 8538

6406531292007. ✔ 816

6406531292008. ✖ 18

Question Number : 20 Question Id : 640653388719 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Multiple Choice Question

How many itemsets have support value greater than or equal to 0.3?

Options :

6406531292009. ✓ 13

6406531292010. ✖ 9

6406531292011. ✖ 21

6406531292012. ✖ 7

Question Number : 21 Question Id : 640653388720 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

What is the probability of observing {Price_Premium = 3, Quality_reputation =3}?

Options :

6406531292013. ✖ 0.33

6406531292014. ✖ 0.55

6406531292015. ✓ 0.4

6406531292016. ✖ 0.3

Question Number : 22 Question Id : 640653388721 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

How many association rules have minimum support of 0.4?

Options :

6406531292017. ✓ 12

6406531292018. ✖ 25

6406531292019. ✖ 38

6406531292020. ✖ 15